Myungsin Cho

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Appointment

Aug 2025 – **Ritt Assistant Professor**, Columbia University

Education

Jul 2025 **Ph.D. in Mathematics**, Indiana University
- Advisor: **Prof. Michael Mandell**

Aug 2018 **M.Sc. in Mathematics**, Seoul National University
- Advisor: **Prof. Otto van Koert**

Aug 2015 **B.Sc. in Engineering**, Korea Aerospace University

Research Interests

Stable homotopy theory, algebraic K-theory, and equivariant algebra, with particular interests in duality, trace methods, and connections to number theory.

Publications

- **K-theoretic Tate-Poitou duality at prime 2**, in *Advances in Mathematics*, 477 (2025), Paper No. 110370, 26 pp.

Preprints

- **On homological Real trace methods** with T. Gerhardt, L. Keenan, J. Moreno, J.D. Quigley. *arXiv:*2608.04297, 2026. submitted
- **Realizing Compatible Pairs of Transfer Systems by Combinatorial N_∞-Operads** with D. Chan, D. Mehrle, P. Sanchez Ocal, A. Osorno, B. Szczesny, and P. Verdugo, Preprint, *arXiv:*2510.26047, 2025. submitted
- **Structural invariance of Green-Griffiths-Demailly Thresholds on compact complex orbifolds** with G. Cho. *arXiv:*2511.05830, 2025. submitted

In preparations

- **Grothendieck-Witt theory and Tate-Poitou duality** with L. Yang.
- **Topological graded Tambara functors** with N. Wisdom.

Honors & Awards

2025 Outstanding Thesis Award, Indiana University
2024–2025 College of Arts and Sciences Dissertation Research Fellowship, Indiana University
2019 Robert E. Weber Memorial Award, Indiana University

2016 Lecture and Research Scholarship, Seoul National University

Talks

Oct 2026 Atlantic Meeting on Topology, Representation theory, and K-theory, University of Pennsylvania

Apr 2026 Columbia University, Algebraic Topology Seminar

Mar 2026 Indiana University, Topology Seminar

Jan 2026 Joint Mathematics Meetings, Special Session on New Voices in Homotopy Theory

Oct 2025 Texas State University, Topology Seminar

Jul 2025 Scissors Congruence and K-theory

Apr 2025 Indiana University, Algebra Seminar

Mar 2025 AMS 2025 Spring Central Sectional Meeting, Special Session on Homotopy Theory and Algebraic K-Theory

Oct 2024 Columbia University, Algebraic Topology Seminar

Oct 2024 University of Virginia, Topology Seminar

Sep 2024 Indiana University, Topology Seminar

Sep 2024 Ohio State University, Homotopy Seminar

Aug 2024 MathFest 2024, Contributed Session: Advances in Algebraic Topology, Indianapolis

Jan 2024 FRG Virtual Seminar

Nov 2023 BUGCAT Conference, Binghamton University

Jun 2023 Scissors Congruence, Algebraic K-Theory, and Trace Methods, Indiana University

Teaching

Columbia University

Introduction to Modern Algebra I, (S2027, scheduled)

Analysis and Optimization (F2026)

Linear Algebra (Su2026)

Analysis and Optimization (S2026)

Analysis and Optimization (F2025)

Indiana University

Calculus 1 (F2022)

Mathematics of Decision and Beauty (Co-instructor, Su2020, F2020, S2021)

Mentoring

Directed Reading Program (Indiana University)

Fall 2023 *Exponential Law in Vector Spaces - A Glimpse of the Adjoint Isomorphism Theorem.* Book: J. Rotman, *An Introduction to Homological Algebra.*

Spring 2023 *Simplicial Homotopy Theory.* Book: P. Goerss and R. Jardine, *Simplicial Homotopy Theory.*

Fall 2022 *A Ring Structure on Vector Bundles.* Book: M. Aguilar, S. Gitler, C. Prieto, *Algebraic Topology from a Homotopical Viewpoint.*

Service and Organizational Activities

2025–2027 Algebraic Topology Seminar, Columbia University

2023–2025 Topology Seminar, Indiana University

Apr 2025 Graduate Student Topology and Geometry Conference 2025, Indiana University

Fall 2023 Reading Seminar on Equivariant Stable Homotopy Theory, Indiana University

May 2023 Reading Seminar on Stable Homotopy Theory, Indiana University

2021 Graduate Student Homotopy Theory Seminar, Indiana University

2015–2018 Mathemaniac (Graduate Student Biannual Seminar), Seoul National University

References

- **Michael Mandell**
  Department of Mathematics, Indiana University
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- **Andrew Blumberg**
  Department of Mathematics, Columbia University
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- **Ayelet Lindenstrauss**
  Department of Mathematics, Indiana University
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- **Vladimir Eiderman** (teaching)
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- **Soren Galatius** (teaching)
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